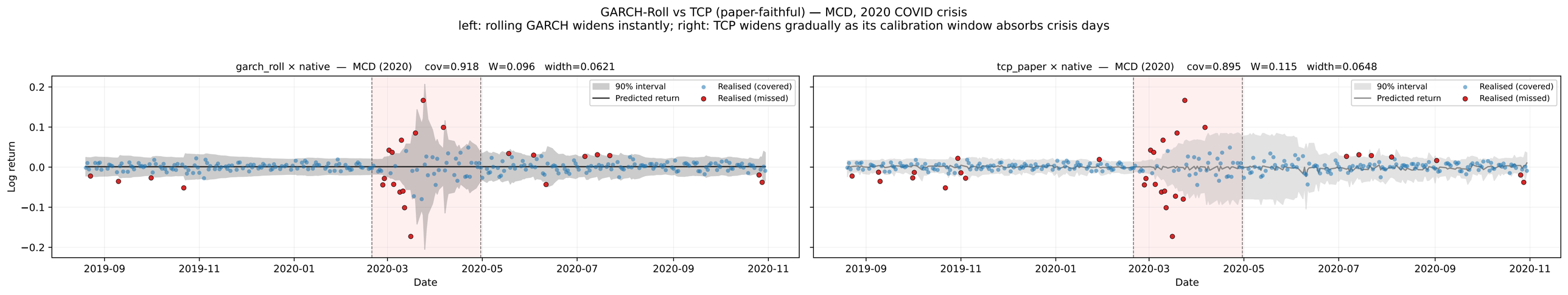
GARCH-Roll vs TCP (paper-faithful) — MCD, 2020 COVID crisis
left: rolling GARCH widens instantly; right: TCP widens gradually as its calibration window absorbs crisis days
garch_roll × native — MCD (2020) cov=0.918 W=0.096 width=0.0621
tcp_paper × native — MCD (2020) cov=0.895 W=0.115 width=0.0648
90% interval
Predicted return
Realised (covered)
Realised (missed)
Log return
Date
0.2
0.1
0.0
−0.1
−0.2
2019-09
2019-11
2020-01
2020-03
2020-05
2020-07
2020-09
2020-11